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ADI DC2703A-A-KIT battery charger with MPPT and telemetry

The DC2703A-A-KIT from Analog Devices is a reference design for a Li-ion battery charger with maximum power point tracking (MPPT) and telemetry. It employs IC LT8491 for charging and management of Li-ion battery and DC1613A, a USB to I2C controller, for enabling communication.

The board is configured for a wide range of input voltages from 17V to 54V and is suitable for various power sources, which include various DC sources like solar panels with 36 to 72 cells with a total power of up to 200W. The module has both buck and boost configurations, thus it can take input voltage below, above, or equal to the battery voltage. Apart from the Li-ion batteries, the charger is suitable for various other battery chemistries, including the flooded and sealed lead-acid batteries.

The datasheet indicates that the charger is up to 97% efficient when charging with a current of 6A. The efficiency of the charger decreases as the charging current increases.

The one feature which sets it apart from any other product is the MPPT for solar-powered applications. The on-chip logic of LT8491 IC enables true MPPT function, which continuously tracks the maximum power point at all times. In order to select the maximum value on the power curve, it periodically sweeps the input panel voltage.

The LT8491 includes a slave I2C compatible interface for digital control of the charger settings and digital readouts of charger telemetry and status. The module is compatible with an MS Windows based GUI application called simpleLT8491, which enables the user to monitor

and control the telemetry and status data of the charger and battery pack in real time.

The DC2703A-A-KIT is capable of supporting higher current and voltage for different cell configuration by replacing the input side MOS-FETs and capacitors with ones with higher ratings.

Important components.

U1 - ADI LT8491 battery charge controller IC

U2 - ADI LT1636CS8#PBF op-amp

U3 - ADI LTC4359CDCB ideal diode controller, 6-pin DFN EP

U4 - 24LC025-I/ST memory EEPROM, 2kb

Features of DC2703A-A-KIT.

- Operation from solar panel or DC supply
- VIN range: 17V to 54V
- VBAT range: 1.3V to 80V
- Single inductor allows VIN above, below, or equal to VBAT
- Automatic MPPT for solar powered charging
- Automatic temperature compensation
- I2C telemetry and configuration
- Remote sensing of battery voltage

This charger reference design can be used as a solar charger or a standalone charger for multiple cell chemistries, which include Li-ion, LiPo, and SLA.

More details about the reference design can be viewed by scanning the QR code.



MAXREFDES1219 - a USB BC 1.2 compliant design

The MAXREFDES1219 is an integrated charger design for the 1-cell Li + battery. The module is an optimum charger for both charging and monitoring. For charging management, it employs MAX14746 IC, which is a USB charger that integrates a

charger detector, linear regulator, and smart power selector to provide up to 2A of charging current.

The module offers features such as USB charger detection and battery cell monitoring, which includes state of charge, open circuit voltage, temperature, current, etc. The reference design is compliant with USB Battery Charging Rev1.2, which is capable of detecting different battery charging methods, including standard downstream port, charging downstream port, and dedicated charging port. The devices also detect common proprietary charge adaptors, including those from Apple 1.

This reference design uses Maxim Integrated's MAX17260 fuel gauge IC, which employs the Model-Gauge m5 algorithm that offers high estimation accuracy and has a low power consumption. The charger enables the user to set the charge termination voltage between 4.05V and 4.6V.

The MAXREFDES1219 can be configured quickly and easily by using the GUI software provided by the company. It requires a MAX-32660EVSYS breakout board for connecting it with the PC. The MAX32660 module can enable read and write control of the two MAX14746/MAX17260 registers through the I2C interface.

The GUI's demo view window displays the status and enables the control of both the charger and the fuel gauge monitor. The software is easy to use and automatically detects the hardware on making the connection.

Important components.

U1 - MAX14746 charging IC

U2 - MAX17260 fuel gauge IC

Features.

- USB Battery Charging Rev. 1.2 compliant
- Capable of detecting proprietary charger, such as Apple 1
- Simple GUI for monitoring and